

K_a and K_b calculations (HL)

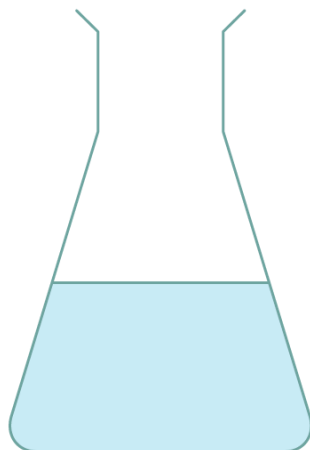
Higher level (HL)

Learning outcomes

By the end of this section you should be able to solve problems involving K_a and K_b values.

Use **Interactive 1** to remind yourself about K_a and pH.

Look at the relationship between K_a and pH.



0.10 M
hydrochloric acid
HCl
100% ionised
 $K_a = 1000$

↺ Turn

Card 1 of 3



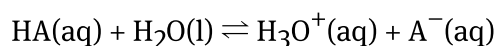
Interactive 1. Relating K_a and pH.

Now that you understand the difference between strong and weak acids, can you calculate the pH of a weak acid in the same way in which we calculated pH of a strong acid? Why? What is different?

We will revisit relationships that we have studied throughout this subtopic and put them in mathematical relationships. We will start with associating numbers to the relative strength of acids and bases in conjugate acid–base pairs.

Conjugate acid—base pairs

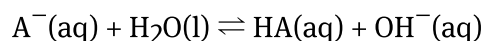
For an acid:



and

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

For its conjugate base A[−]:



and

$$K_b = \frac{[\text{OH}^-][\text{HA}]}{[\text{A}^-]}$$

If K_a and K_b are multiplied then:

$$K_a \times K_b = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} \times \frac{[\text{OH}^-][\text{HA}]}{[\text{A}^-]}$$

which simplifies to:

$$K_a \times K_b = [\text{H}_3\text{O}^+][\text{OH}^-]$$

But we know from section R3.1.4–5a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)) that:

$$[\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

Therefore:

$$K_a \times K_b = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

Taking the logarithm of both sides, we obtain:

$$\text{p}K_a + \text{p}K_b = 14$$

Therefore, for a conjugate acid–base pair:

- $K_a \times K_b = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
- $\text{p}K_a + \text{p}K_b = 14$

We can conclude that the stronger the acid and therefore the higher its K_a value, then the lower the K_b value of the conjugate base, and the weaker the conjugate base.

Study skills

Remember for any conjugate acid–base pair:

- $K_a \times K_b = K_w$
- $\text{p}K_a + \text{p}K_b = \text{p}K_w$
- $\text{p}K_a + \text{p}K_b = 14.00$ at 298 K

Calculations of pH for weak acids and bases

When calculating the pH of strong acids and bases we assume full dissociation into ions in aqueous solution. Because of this, we can use the initial concentration of the acid or base to calculate the pH or pOH as seen in [section R3.1.4–5a](#)

([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

Weak acids and weak bases only partially dissociate in solution, which means that unlike for strong acids and bases, we cannot determine the $[H^+]$ or $[OH^-]$ in solution from their initial concentrations.

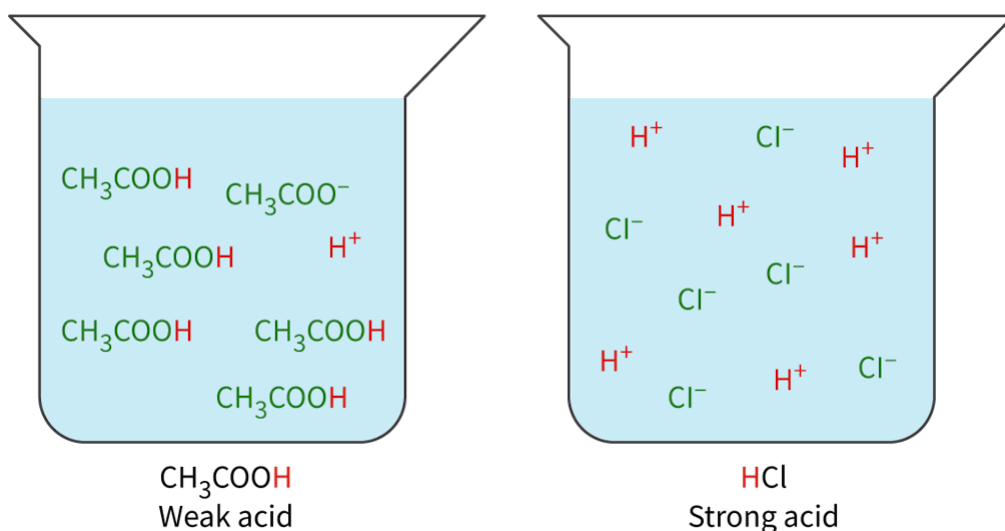


Figure 1. Dissociation of weak vs. strong acid.

 More information for figure 1

The image shows two beakers, each containing a liquid with dissociated molecules. The left beaker is labeled with the chemical notation for weak acid, acetic acid (CH_3COOH). Inside the left beaker, several undissociated acetic acid molecules (CH_3COOH) are present alongside a few dissociated ions, acetate (CH_3COO^-) and hydrogen (H^+). This illustrates partial dissociation characteristic of weak acids.

The right beaker is labeled as containing a strong acid, hydrochloric acid (HCl). In this beaker, ions fully dissociated into hydrogen (H^+) and chloride (Cl^-) are depicted, indicating complete dissociation characteristic of strong acids. This comparison visually demonstrates the difference between the dissociation of weak and strong acids in aqueous solutions.

[Generated by AI]

As seen in **Figure 1**, for weak acids, only a few molecules dissociate. The number of dissociated molecules is very small. Therefore, if the acid is weak, the equilibrium concentration of the undissociated acid molecules can be taken as the initial concentration of the acid, $[HA]_{eq} = [HA]_{initial}$. The assumption here is that very few

acid molecules have actually dissociated. A similar assumption is made for the concentration of the undissociated molecules of a weak base at equilibrium; namely, that the equilibrium concentration of the weak base is approximately equal to the initial concentration of the weak base, $[B]_{\text{eq}} = [B]_{\text{initial}}$.

Worked example 1

A $0.0100 \text{ mol dm}^{-3}$ solution of a weak acid HA at 298 K has a pH of 4.17. Calculate the K_{a} of the acid.

First, determine the $[H^+]$ from pH.

$$\begin{aligned} [H^+] &= 10^{-\text{pH}} \\ &= 10^{-4.17} \\ &= 6.76 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

The K_{a} equation is:

$$K_{\text{a}} = \frac{[H^+][A^-]}{[HA]}$$

Let x be $[H^+]$ at equilibrium.

Using an ICE table gives:

	HA (aq) (mol dm^{-3})	H^+ (ac) (mol dm^{-3})
Initial concentration	0.0100	0
Change in concentration	$-x$	x
Equilibrium concentration	$0.0100 - 6.76 \times 10^{-5}$	$6.76 \times 10^{-5} \text{ dm}^{-3}$

Note that the ratio of $[H^+]$ to $[A^-]$ is 1:1, therefore:

$$\begin{aligned} x &= [H^+] \\ &= [A^-] \\ &= 6.76 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

and

$$[HA] = 0.0100 - 6.76 \times 10^{-5}$$

When rounding to 4 significant figures:

$$0.0100 - 6.76 \times 10^{-5} \cong 0.0100 \text{ mol dm}^{-3}$$

therefore we can omit the $-x$ term in the calculations below. This is an assumption and approximation that can almost always be applied in this course especially because K_a values are relatively small and therefore x is a small value.

Substitute the values into the K_a equation:

$$\begin{aligned} K_a &= \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \\ &= \frac{(6.76 \times 10^{-5})^2}{0.01} \\ &= 4.60 \times 10^{-7} \text{ mol dm}^{-3} \end{aligned}$$

Worked example 2

The value of the K_a for ethanoic acid, CH_3COOH , is 1.8×10^{-5} at 25°C . Calculate the pH of a 0.10 mol dm^{-3} solution of CH_3COOH .

Ethanoic acid dissociates according to the equation:



The K_a equation is:

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

Let x be $[\text{H}^+]$ at equilibrium.

Using an ICE table gives:

	$\text{CH}_3\text{COOH}(\text{aq})$ (mol dm^{-3})	$\text{H}^+(\text{aq})$ (mol dm^{-3})
Initial concentration	0.10	0
Change in concentration	$-x$	x
Equilibrium concentration	$0.10 - x$	x

Remember the assumption:

Because of the very small value of the K_a , we can make the assumption that $0.10 - x \approx 0.10$

For this acid $[H^+] = [CH_3COO^-] = x$, therefore:

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$$1.8 \times 10^{-5} = \frac{x^2}{0.10}$$

$$\begin{aligned} x^2 &= 1.8 \times 10^{-5} \times 0.10 \\ &= 1.8 \times 10^{-6} \end{aligned}$$

$$x = 1.3 \times 10^{-3}$$

$$[H^+] = 1.3 \times 10^{-3} \text{ mol dm}^{-3}$$

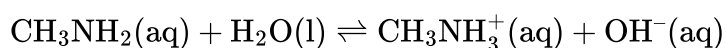
$$\text{pH} = -\log[H^+]$$

$$\begin{aligned} \text{pH} &= -\log(1.3 \times 10^{-3}) \\ &= 2.89 \end{aligned}$$

Worked example 3

The value of the K_b for methylamine (IUPAC name methanamine), CH_3NH_2 , is 4.6×10^{-4} at 25°C. Calculate the pH of a 0.10 mol dm^{-3} solution of CH_3NH_2 .

Methylamine reacts with water as shown in the equation:



The K_b equation is:

$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2]}$$

Let x be $[H^+]$ at equilibrium.

Because of the very small value of the K_b , we can make the assumption that the $[CH_3NH_2]_{eq} = [CH_3NH_2]_{initial}$.

Using an ICE table gives:

	CH ₃ NH ₂ (aq) (mol dm ⁻³)	OH ⁻ (mol dm ⁻³)
[] initial (mol dm ⁻³)	0.10	0
Change in [] (mol dm ⁻³)	-x	x
[] eq (mol dm ⁻³)	0.10 - x	x

For this base:

$$[\text{OH}^-] = [\text{CH}_3\text{NH}_3^+] \\ = x$$

Because of the weak nature of the base, K_b is small, and we can make the assumption that $0.10 - x \approx 0.10$, therefore:

$$4.6 \times 10^{-4} = \frac{x^2}{0.10}$$

$$x^2 = 4.6 \times 10^{-4} \times 0.10$$

$$x = 6.8 \times 10^{-3}$$

$$[\text{OH}^-] = 6.8 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(6.8 \times 10^{-3})$$

$$\text{pOH} = 2.17$$

At 25 °C,

$$\text{pH} + \text{pOH} = 14.00$$

$$\text{pH} = 14.00 - 2.17 \\ = 11.83$$

Making connections

How can we simplify calculations when equilibrium constants K_a and K_b are very small? (see [subtopic R2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/))

Activity

- **IB learner profile attribute:** Knowledgeable:
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions

1. Research the K_a of at least six acids then the K_b of six bases.
2. Create a spreadsheet like the one below.
3. Calculate the K_b of the conjugate base for the acids, and the K_a of conjugate acid for the bases.
4. Calculate the corresponding pK_a and the pK_b .

Acid/ base	K_a/K_b value	K_b/K_a of conjugate base/acid